

Assignment 2

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In this part, we analyze a dataset containing words from the book *Alice in Wonderland*. The objective is to study the distribution of the number of extra letters in each word and investigate whether a Poisson distribution provides a reasonable model for the data.

Table 1: First 6 rows of the dataset (alice.csv)

word	wordlength
Adventures	10
in	2
Wonderland	10
ALICES	6
ADVENTURES	10
IN	2

Since we want to model the data using a Poisson distribution, the response variable must start from zero. However, there are no words with length zero in the dataset. To overcome this issue, we apply a simple transformation and define a new variable called *extraletters*, which is obtained by subtracting one from the word length

```
extraletters = alice$wordlength - 1
```

This transformation shifts the support of the variable to start at 0, which matches the support of a Poisson distribution.

To examine the distribution of the variable *extraletters*, we compute its relative frequency and visualize it using a bar plot. This allows us to assess whether the data resemble a Poisson distribution.

Summary of the transformed variable

Length	Class	Mode
0	NULL	NULL

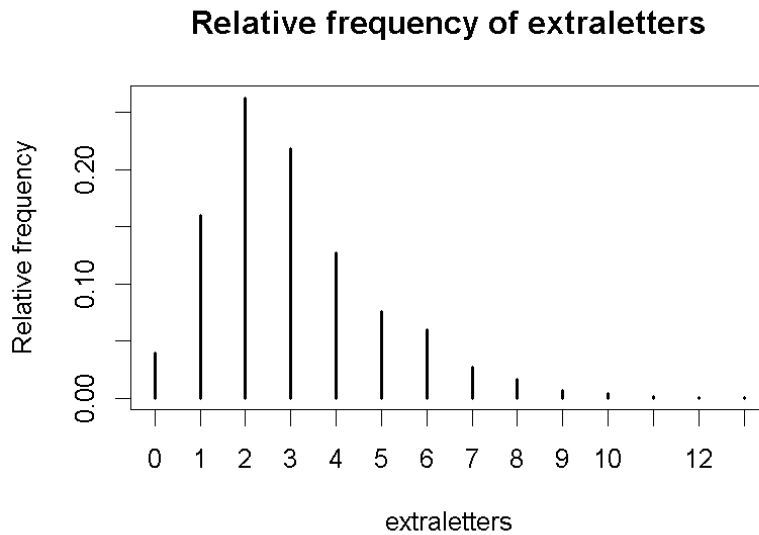
Estimation of the Poisson parameter

For a Poisson random variable, the parameter λ is equal to the expected value:

$$\lambda = E(X)$$

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i$$

Relative frequency of extraletters



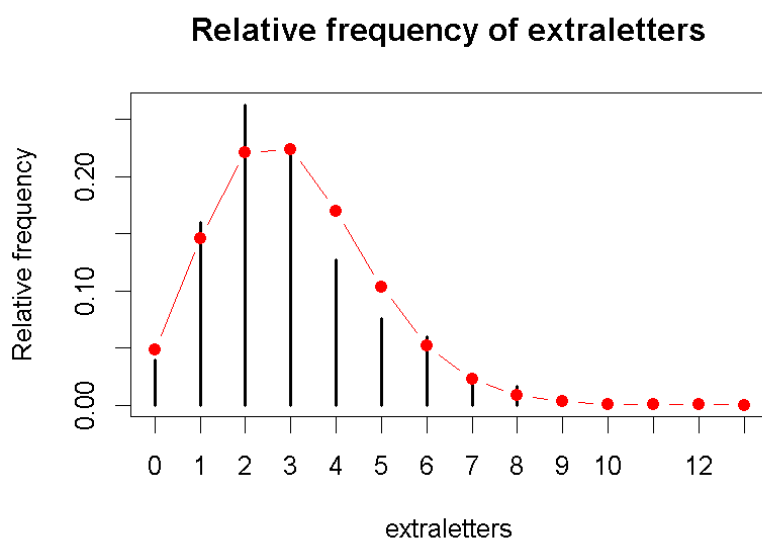
The figure shows the relative frequency of the variable *extraletters*. The distribution has a shape that is consistent with a Poisson distribution.

Poisson distribution fit

To assess whether a Poisson model is reasonable for the variable *extraletters*, we use the estimate $\hat{\lambda}$ obtained from the sample mean. Under a Poisson model, the probability mass function is

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Using $\hat{\lambda}$, the fitted Poisson distribution can be compared to the empirical relative frequencies. Overall, the Poisson model provides a reasonable approximation to the observed distribution, although small discrepancies may appear for larger values.



The Poisson distribution fitted with $\hat{\lambda}$ is compared with the observed relative frequencies.

Probability that a word is 6 letters long or longer

Recall that $X = \text{wordlength} - 1$. Therefore, $\text{wordlength} \geq 6$ is equivalent to $X \geq 5$.

$$P(\text{wordlength} \geq 6) = P(X \geq 5) = 1 - P(X \leq 4).$$

Using the fitted Poisson model with parameter $\hat{\lambda}$, we obtain

$$P(\text{wordlength} \geq 6) \approx 0.1908.$$

Checking the Poisson assumption (mean and variance)

For a Poisson distribution, the mean and variance are equal:

$$E(x) = \text{var}(x) = \lambda$$

A simple diagnostic is the dispersion ratio:

$$\frac{\text{var}(x)}{E(x)}$$

In a Poisson distribution, the mean and the variance are equal, implying that a variance-to-mean ratio close to 1 indicates Poisson-like behavior. If the ratio is greater than 1, the data exhibit overdispersion, whereas a ratio smaller than 1 indicates underdispersion. In our data, the variance-to-mean ratio is approximately 1.21, which is close to the value expected under a Poisson model. Therefore, the Poisson distribution can be considered an acceptable approximation for the data.

Bootstrap and t-test confidence intervals

For this analysis, we assume that only a sample of the data is available. A random sample of 500 observations was drawn from the data after setting a seed to ensure reproducibility.

Using this sample, a bootstrap procedure with 1000 resamples was applied to estimate a 95% confidence interval for the mean word length. The resulting bootstrap confidence interval is

$$[2.82, 3.15].$$

For comparison, a one-sample t-test was used to compute a 95% confidence interval for the mean based on the same sample. The t-test confidence interval is

$$[2.82, 3.13].$$

Conclusion

The Poisson model fits the data reasonably well. The estimated parameter is stable, and the main assumptions of the Poisson distribution are approximately satisfied. Therefore, the Poisson distribution can be used as a suitable model for the word length data.

Linear Regression

We fitted a multiple linear regression model to explain the starting price of homes. The explanatory variables included area, number of rooms, region, housing type, and whether the apartment has a balcony.

The purpose of this model is to investigate how these characteristics are related to housing prices and to evaluate whether a linear model is appropriate.

The estimated multiple linear regression model can be written as:

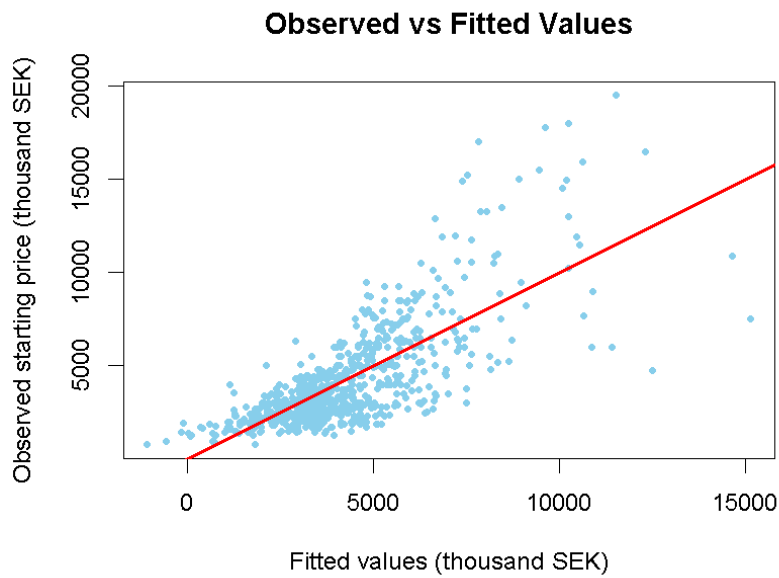
$$\begin{aligned}\text{STARTING_PRICE} = & -413966 \\ & + 332722 I(\text{REGION} = \text{Northwest}) \\ & - 863092 I(\text{REGION} = \text{Southeast}) \\ & + 1696293 I(\text{REGION} = \text{Stockholm}) \\ & - 1586230 I(\text{REGION} = \text{West}) \\ & - 1225422 I(\text{TYPE} = \text{Terrace}) \\ & - 37996 I(\text{TYPE} = \text{Villa}) \\ & - 141083 I(\text{BALCONY} = \text{Yes}) \\ & - 159293 \text{ ROOMS} \\ & + 63625 \text{ AREA} + \varepsilon\end{aligned}$$

Here, $I(\cdot)$ denotes an indicator variable that equals 1 if the condition is satisfied and 0 otherwise. The reference categories are REGION = Northeast, TYPE = Apartment, and BALCONY = No.

The region, housing type, and balcony variables are included in the model as dummy variables. The reference categories are REGION = Northeast, TYPE = Apartment, and BALCONY = No.

The coefficient of determination for the model is $R^2 = 0.55$. This means that about 55% of the variation in starting prices can be explained by the variables included in the model. The remaining variation is due to other factors that are not included in the model.

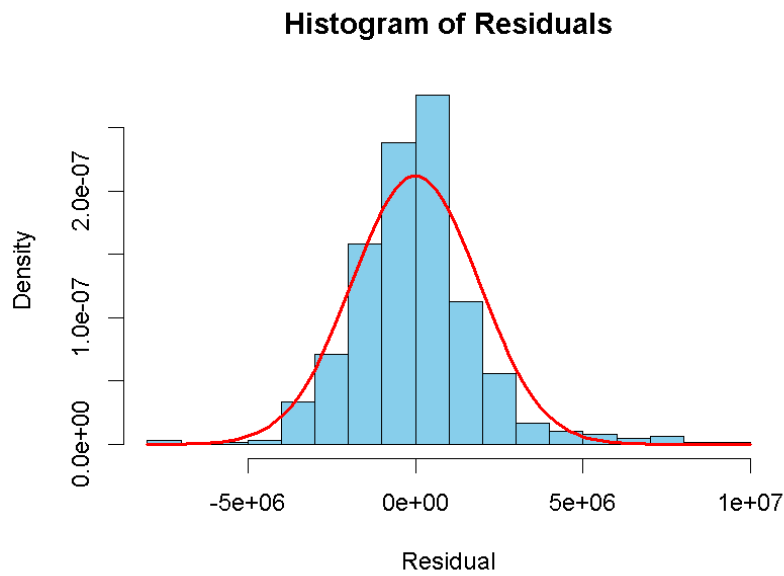
To evaluate how well the model fits the data, we compare the observed starting prices with the fitted values from the model.



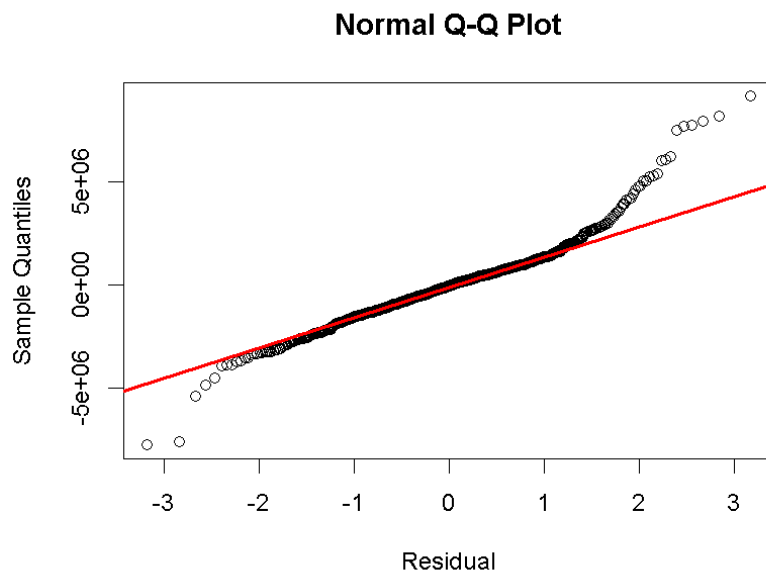
The plot shows a large spread around the line, especially for higher prices. This indicates that the model does not predict prices equally well across all price levels. To better understand how well the model fits the data, we use diagnostic plots.



The residuals versus fitted values plot shows that the residuals are more spread out for larger fitted values. This suggests that the variance of the residuals is not constant.



The histogram of the residuals is right-skewed and does not closely follow a normal distribution. This indicates that the normality assumption is not fully satisfied.



From the Q-Q plot, we see that the points deviate from the reference line, especially in the tails. This also suggests departures from normality.

Based on the diagnostic plots, the linear regression assumptions are not fully met.

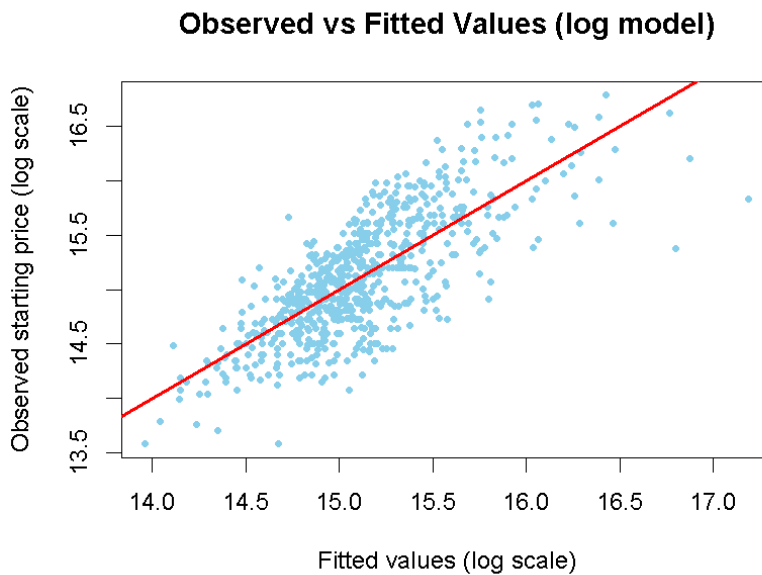
Log-transformed model

To improve the model fit, we apply a logarithmic transformation to the starting price and fit the regression model again using the same explanatory variables.

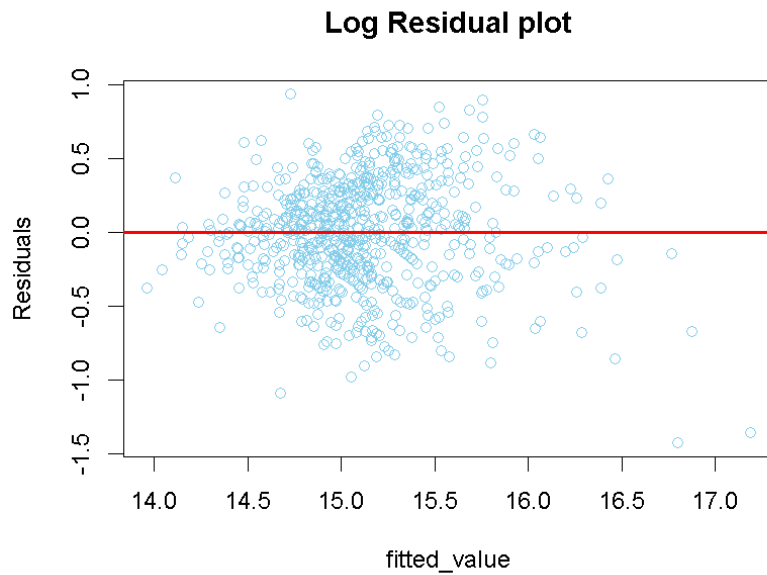
$$\begin{aligned}\log(\text{STARTING_PRICE}) = & 14.162 \\ & + 0.145 I(\text{REGION} = \text{Northwest}) \\ & - 0.173 I(\text{REGION} = \text{Southeast}) \\ & + 0.380 I(\text{REGION} = \text{Stockholm}) \\ & - 0.391 I(\text{REGION} = \text{West}) \\ & - 0.054 I(\text{TYPE} = \text{Terrace}) \\ & + 0.116 I(\text{TYPE} = \text{Villa}) \\ & - 0.024 I(\text{BALCONY} = \text{Yes}) \\ & + 0.022 \text{ROOMS} \\ & + 0.010 \text{AREA} + \varepsilon\end{aligned}$$

The model is estimated on the logarithmic scale, which means that the coefficients can be interpreted as approximate percentage changes in the starting price.

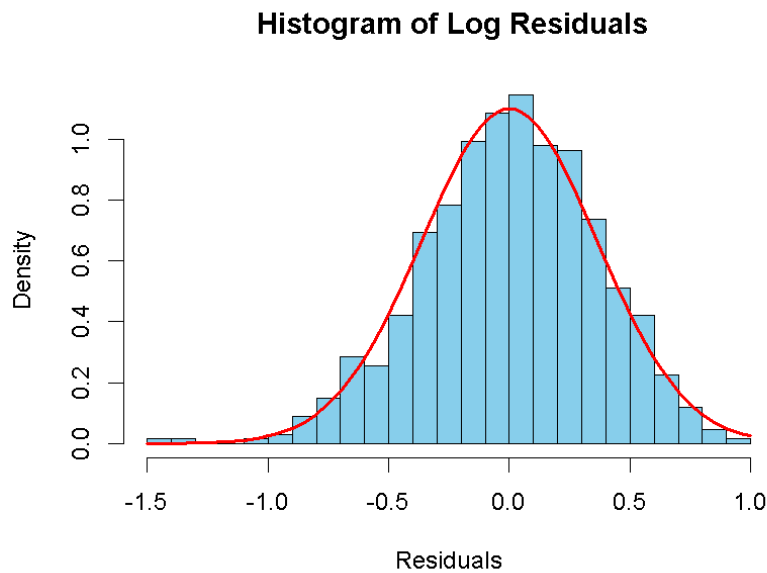
For the log-transformed model, the coefficient of determination is $R^2 = 0.58$, which is slightly higher than for the original model. This indicates that the log-transformed model explains a larger proportion of the variation in housing prices.



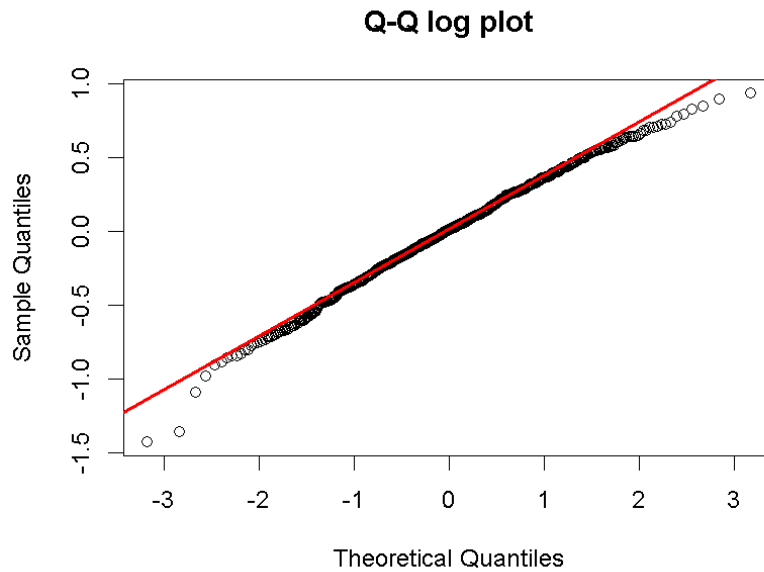
We now examine the diagnostic plots for the log-transformed model.



The residuals versus fitted values plot shows a more even spread of residuals across the fitted values. The variance appears more constant, and the points do not follow any specific pattern.



The histogram of residuals appears more symmetric compared to the original model, suggesting improved normality. However, a slight right skew can still be observed.



The Q-Q plot shows that the residuals are closer to the reference line, although some deviations remain in the tails.

Overall, the log-transformed model provides a better fit than the original linear regression model and better satisfies the regression assumptions.

Backward Elimination

We start the backward elimination procedure from the full log-transformed model.

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.1624959  0.0551827 256.647 < 2e-16 ***
REGIONNorthwest  0.1447460  0.0554559   2.610  0.00926 **
REGIONSoutheast -0.1731673  0.0555978  -3.115  0.00192 **
REGIONStockholm  0.3803765  0.0412732   9.216 < 2e-16 ***
REGIONWest     -0.3912113  0.0547045  -7.151 2.31e-12 ***
TYPETerrace    -0.0544399  0.0668107  -0.815  0.41546
TYPEVilla      0.1157456  0.0709637   1.631  0.10336
BALCONYYes     -0.0242612  0.0325876  -0.744  0.45685
ROOMS          0.0219631  0.0243096   0.903  0.36661
AREA           0.0099640  0.0009418  10.579 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.365 on 654 degrees of freedom
Multiple R-squared:  0.5794,    Adjusted R-squared:  0.5736
F-statistic: 100.1 on 9 and 654 DF,  p-value: < 2.2e-16

```

The summary of this model shows that the variable BALCONY has a high p-value and is not statistically significant.

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.1490468  0.0521243 271.448 < 2e-16 ***
REGIONNorthwest  0.1426474  0.0553654   2.576  0.01020 *
REGIONSoutheast -0.1735439  0.0555766  -3.123  0.00187 **
REGIONStockholm  0.3815961  0.0412267   9.256 < 2e-16 ***
REGIONWest     -0.3887026  0.0545820  -7.121 2.82e-12 ***
TYPETerrace    -0.0385743  0.0632993  -0.609  0.54247
TYPEVilla      0.1343822  0.0663793   2.024  0.04333 *
ROOMS          0.0219992  0.0243013   0.905  0.36566
AREA           0.0099040  0.0009381  10.558 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3648 on 655 degrees of freedom
Multiple R-squared:  0.579,    Adjusted R-squared:  0.5739
F-statistic: 112.6 on 8 and 655 DF,  p-value: < 2.2e-16

```

After removing BALCONY from the model, the adjusted R^2 increases slightly and the residual standard error decreases. This shows that removing BALCONY does not weaken the model and slightly improves its performance.

In the next step of the backward elimination process, we look at whether the variable ROOMS should be removed. In the previous model, ROOMS has a high p-value, which means it is not statistically significant.

Because AREA already represents the size of the apartment, ROOMS does not provide much additional information.

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.1577629  0.0512203  276.409 < 2e-16 ***
REGIONNorthwest  0.1415406  0.0553443   2.557  0.01077 *
REGIONSoutheast -0.1736838  0.0555687  -3.126  0.00185 **
REGIONStockholm  0.3842404  0.0411174   9.345 < 2e-16 ***
REGIONWest     -0.3913482  0.0544962  -7.181  1.88e-12 ***
TYPETerrace     -0.0238290  0.0611593  -0.390  0.69694
TYPEVilla       0.1553754  0.0621882   2.498  0.01272 *
AREA           0.0106124  0.0005171  20.523 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3648 on 656 degrees of freedom
Multiple R-squared:  0.5785,    Adjusted R-squared:  0.574
F-statistic: 128.6 on 7 and 656 DF,  p-value: < 2.2e-16

```

After removing ROOMS from the model, the adjusted R^2 stays almost the same, and the residual standard error also remains at a similar level.

Therefore, ROOMS is removed from the model.

After removing ROOMS, we also consider removing the variable TYPE. However, when TYPE is excluded from the model, the adjusted R^2 decreases and the residual standard error increases.

This indicates that TYPE contains relevant information for explaining housing prices. Therefore, TYPE should not be removed. As a result, the backward elimination procedure stops with a final model that includes REGION, TYPE, and AREA.

Variance Inflation Factor (VIF)

We use the Variance Inflation Factor (VIF) to check for multicollinearity among the explanatory variables in the final model.

The variance inflation factor for a variable X_j is defined as

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination obtained by regressing X_j on all other explanatory variables.

As a rule of thumb, VIF values can be interpreted as follows: - $\text{VIF} < 5$: no serious multicollinearity - VIF between 5 and 10: moderate multicollinearity, which may require attention depending on the purpose of the model - $\text{VIF} > 10$: severe multicollinearity

The VIF values for the final model are shown below:

- REGION: $\text{GVIF}^{1/(2 \cdot \text{Df})} = 1.02$
- TYPE: $\text{GVIF}^{1/(2 \cdot \text{Df})} = 1.19$
- AREA: $\text{VIF} = 1.39$

All VIF values are well below 5 and are close to 1. This indicates that there is no problematic multicollinearity among the explanatory variables in the model.

Hypothesis Test for the Coefficient of AREA

We test whether the coefficient of AREA is different from zero in the model

$\log(\text{STARTING_PRICE}) \sim \text{AREA}$.

The hypotheses are:

$$H_0 : \beta_{\text{AREA}} = 0 \quad H_1 : \beta_{\text{AREA}} \neq 0$$

We use a 5% significance level ($\alpha = 0.05$). The test statistic is:

$$t = \frac{\hat{\beta}_{\text{AREA}}}{SE(\hat{\beta}_{\text{AREA}})} = \frac{0.0112311}{0.0003994} \approx 28.12$$

This matches the t value reported in the R output ($t = 28.120$). With $df = 658$, the two-sided critical value is approximately $t_{0.025, 658} \approx 1.96$. We reject H_0 if $|t| > 1.96$.

Since $|t| = 28.12 > 1.96$, we reject the null hypothesis. Therefore, AREA has a statistically significant effect on the starting price when controlling other variable.

Prediction Interval for the Starting Price

Using the final log-transformed regression model, we calculate a 95% prediction interval for the starting price of a home with the following characteristics:

```
prediction_data <- data.frame( AREA = 50, ROOMS = 2, REGION = "Stockholm", TYPE
= "Apartment", BALCONY = "Yes" )
```

The prediction interval is first calculated on the logarithmic scale using the `predict()` function.

```
pred <- predict(log_model2, newdata = prediction_data, interval = "prediction", level =
0.95)
```

To obtain the prediction interval on the original price scale, the exponential function is applied to the fitted values and interval limits.

```
fit      lwr      upr
1 3515263 1715359 7203785
```

This means that, for a home with the given characteristics, we expect the starting price of an individual property to fall within this interval with 95% probability. Since housing prices vary a lot in practice, a wide prediction interval is not surprising and seems reasonable.

Confidence Interval (Mean Starting Price)

In this question, we compute a 95% confidence interval for the *average* starting price of homes with the given characteristics.

```
conf <- predict(log_model2, newdata = prediction_data, interval = "confidence", level = 0.95)
```

The estimated average starting price is 3,515,263 SEK. The 95% confidence interval for the average starting price is [3,373,341 SEK, 3,663,156 SEK].

This means that, with 95% confidence, the true average starting price for homes with these characteristics lies within this interval.

This confidence interval is much narrower than the prediction interval in Question 6, because it describes uncertainty about the *mean* price, not the price of a single home.